

Mehmet Arda Hakbilen

Undergraduate Researcher, LLM & Efficient Sequence Architectures

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Undergraduate researcher working on the architectures currently competing with attention inside LLMs: state-space models (Mamba-family SSD) and associative-memory / delta-rule mixers (DeltaNet-family). Focused on where design choices that transfer cleanly from language modeling stop transferring. Currently building PRISM, a hybrid linear-recurrent backbone combining both mixer families, stress-tested for cross-modal portability. Writes own kernels and own instrumentation instead of relying on library defaults, and documents negative results, including a closed frozen-LLM memory-injection project, with the same rigor as positive ones.

SKILLS

ML / Research: PyTorch, Triton, CUDA, TensorRT, NumPy

Systems: Python, Go, Rust, Linux

Infra / Tooling: FastAPI, Docker, PyPI packaging

Web / Frontend: TypeScript, React, Next.js, WebSocket

RESEARCH & PROJECTS

PRISM, Modality-Portable Hybrid Linear-Recurrent Backbone

Independent research · active

- 12-layer hybrid backbone interleaving Mamba-2-style SSD blocks with Gated Delta Rule blocks at a **3:1 ratio**, the same mixer families used in current efficient-LLM backbones (Mamba-2, Gated DeltaNet). Stress-tested for cross-modal portability without per-modality tuning.
- Both mixers implemented from scratch (Hillis-Steele scan, UT-transform chunked delta rule); numerical equivalence verified against torch.associative_scan and the FLA Triton kernels across **270+ tests** (equivalence, fp64 gradcheck, streaming state-passing, property-based).
- Identical backbone and hyperparameters applied across PTB-XL ECG (primary), Speech Commands, and sequential CIFAR-10 to test cross-modal generalization; target is macro-AUROC parity with xresnet1d101 (~**0.928**). Pipeline validated end-to-end on an RTX 5060 laptop; full paper matrix not yet run.
- Sub-project (MoM): replacing the fixed SSD:GDR ratio with a per-token router that selects the memory primitive per token. First spike-gate attempt did not collapse but underperformed the GDR-only baseline on recall; currently working through mitigations.
- Targeting **ICML 2026 ES-FoMo IV / NeurIPS 2026 ENLSP-VI** workshops.

Trainscope, PyTorch Training Instrumentation Library

Published on PyPI · solo

- Real-time training monitor for catching loss spikes and silent divergence early: Welford online z-score spike detection plus CUSUM-based drift detection.
- FastAPI backend, React/Plotly frontend, Apache Arrow IPC transport. Built out of a need during PRISM's own training runs, with several production-hardening passes since first release.

NOESIS, Frozen-Backbone Vector-Memory Injection

Independent research · closed, Aug 2026

- Tested whether an external vector-memory module could inject new semantic content into a frozen LLM backbone without any task-supervised fine-tuning.
- Result, verified via systematic benchmarking: injection alone does not transfer semantic content into a frozen backbone; task-supervised training is required. Closed and documented as a negative result.

EXPERIENCE

Computer Vision Intern, YourEye Teknoloji

Apr 2026 – Aug 2026 · Ankara, Türkiye · Hybrid

- Built a real-time camera analytics system for object detection and tracking: YOLO-based detection, BoT-SORT tracking, polygon ROI analysis, line-crossing counting.
- Integrated InsightFace for identity recognition into the pipeline; optimized inference with TensorRT/CUDA and handled GPU deployment.

EDUCATION

B.Sc. Software Engineering, Ankara University

Expected 2027

- Undergraduate course work in algorithms, data structures, software architecture, and systems programming.